
Integrable Systems — Exercise Sheet 6

Action–angle variables

Variant: Questions only

Conventions, as in Lectures 2–6: $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$, so that $\dot{f} = \{f, H\}$. An action is $I = \frac{1}{2\pi} \oint p \, dq$ round a cycle traversed in the direction of the motion. For the smooth one-degree-of-freedom wells $H = \frac{1}{2}p^2 + V(q)$ of this sheet, a closed orbit that bounds a region of the plane runs clockwise ($\dot{q} = p > 0$ on its upper branch), and its action is the area of that region over 2π , positive (Lecture 6). With one degree of freedom, $dI/dE = T/2\pi$ and $\nu = dE/dI = 2\pi/T$ (Lecture 6). $K(k)$ and $E(k)$ are the complete elliptic integrals of Lecture 6, in the modulus k ; the energy is E , without an argument.

Exercise 1 (the recipe, on two wells). Lecture 6's recipe: choose the cycle, integrate, invert to get $K(I)$, differentiate to get $\nu(I)$. (i) Let $H = \frac{1}{2}p^2 + \frac{1}{4}q^4$ and $E > 0$. Substituting $q = (4E)^{1/4}x$, show that $I(E) = c E^{3/4}$ with $c = \frac{4}{\pi} \int_0^1 \sqrt{1-x^4} \, dx$. Find $K(I)$ and $\nu(I)$. Does the frequency rise or fall with the action? (ii) Let $H = \frac{1}{2}p^2 + (1 - e^{-q})^2$, the Morse potential, and $0 < E < 1$. Show that the level set $\{H = E\}$ is one closed curve, and that its period is $T(E) = 2\pi/\sqrt{2(1-E)}$. You may use, for $0 < u_- < u_+$,

$$\int_{u_-}^{u_+} \frac{du}{u\sqrt{(u_+ - u)(u - u_-)}} = \frac{\pi}{\sqrt{u_+ u_-}}.$$

(iii) From $T(E)$ and $dI/dE = T/2\pi$, find $I(E)$, $K(I)$ and $\nu(I)$. Compare ν near $I = 0$ with the frequency of small oscillations, and say what happens as $E \rightarrow 1$.

Exercise 2 (the pendulum at small amplitude). For the librating pendulum, $H = \frac{1}{2}p^2 - \cos \theta$ with $-1 < E < 1$, Lecture 6 found $T = 4K(k)$ and $I = \frac{8}{\pi}[E(k) - (1 - k^2)K(k)]$, with $k^2 = (1 + E)/2$. Write $\epsilon = E + 1$, the energy above the bottom of the well. (i) Expand the integrands in powers of $k^2 \sin^2 \psi$ to show that, as $k \rightarrow 0$,

$$K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9k^4}{64} \right) + O(k^6), \quad E(k) = \frac{\pi}{2} \left(1 - \frac{k^2}{4} - \frac{3k^4}{64} \right) + O(k^6).$$

(ii) Deduce $T = 2\pi(1 + \epsilon/8) + O(\epsilon^2)$ and $I = \epsilon + \epsilon^2/16 + O(\epsilon^3)$. Which oscillator is the limit $\epsilon \rightarrow 0$? (iii) Find $K(I)$ to order I^2 , and $\nu(I) = 2\pi/T$ to order I . Check that ν agrees, to that order, with the derivative of the polynomial part of K . Does the frequency rise or fall with the amplitude? (iv) For the swing of amplitude $\theta_0 = \pi/2$, which rises to the height of the pivot, find k , and compute T from $K(1/\sqrt{2}) = 1.85407$. Compare with 2π and with the first-order formula of (ii).

Exercise 3 (the rotating pendulum's action). For $E > 1$ the pendulum $H = \frac{1}{2}p^2 - \cos \theta$ rotates, and Lecture 6 states that on each of the two components $T = 2k K(k)$ and $I = \frac{4}{\pi k} E(k)$, with $k^2 = 2/(1 + E)$. Sheet 1, Exercise 5 derived T . (i) Show that the action of the component with $p > 0$ is $I = \frac{1}{2\pi} \int_0^{2\pi} \sqrt{2(E + \cos \theta)} \, d\theta$, and derive $I = \frac{4}{\pi k} E(k)$. (ii) Show that $dE(k)/dk = (E(k) - K(k))/k$ for $0 < k < 1$, and use it to check that $dI/dE = T/2\pi$. (iii) Show that $I/\sqrt{2E} \rightarrow 1$ as $E \rightarrow \infty$. (iv) Show that $I \rightarrow 4/\pi$ as $E \rightarrow 1$ from above, and that the libration action $I = \frac{8}{\pi}[E(k) - (1 - k^2)K(k)]$ tends to $8/\pi$ as $E \rightarrow 1$ from below. Relate the two limits to the regions that the separatrix $p = \pm 2 \cos(\theta/2)$ bounds.

Exercise 4 (orbit closures and resonance). On the torus $T^2 = \mathbb{R}^2/2\pi\mathbb{Z}^2$ consider $\varphi(t) = \varphi(0) + \nu t$. (i) For $\nu = (1, 2)$ and $\nu = (2, 3)$, show that the orbit is closed and find its period. (ii) For $\nu = (1, \sqrt{2})$, show that the orbit comes arbitrarily close to every point of T^2 : its closure is the whole torus. (iii) For $\nu = (1, 0)$, what is the orbit? State a rule, in terms of ν_1/ν_2 , for when the orbit is closed, and compare it with Lecture 6's definition of a resonant frequency vector. (iv) Let $K(I) = \frac{1}{2}(I_1^2 + I_2^2)$ for $I_1, I_2 > 0$ (two free rotors, with $I_i = p_i$). Find $\nu(I)$, and show that the torus with actions I is resonant exactly when I_1/I_2 is rational. Deduce that the resonant tori are dense in the quadrant, and that they form a set of area zero (a countable union of lines has area zero). What happens instead for the oscillator, $K = \nu_1 I_1 + \nu_2 I_2$?

Exercise 5 (Kepler's polar action). In Lecture 6's Kepler reading, on the level set with $0 < |\ell_3| < \ell$, the polar angle $\theta \in (0, \pi)$ has $p_\theta = \pm \sqrt{\ell^2 - \ell_3^2/\sin^2 \theta}$, and the lecture uses $I_\theta = \frac{1}{2\pi} \oint p_\theta d\theta = \ell - |\ell_3|$. Here the cycle is traversed in the direction in which θ moves, $\dot{\theta} = p_\theta/r^2$. (i) Show that the curve $p_\theta^2 + \ell_3^2/\sin^2 \theta = \ell^2$ is one closed curve, over $\theta_- \leq \theta \leq \theta_+$ with $\sin \theta_\pm = |\ell_3|/\ell$, and that $I_\theta = \frac{1}{\pi} \int_{\theta_-}^{\theta_+} \sqrt{\ell^2 - \ell_3^2/\sin^2 \theta} d\theta$. (ii) Substitute $\cos \theta = u_0 \cos s$, $s \in [0, \pi]$, with $u_0 = \sqrt{1 - \ell_3^2/\ell^2}$, to show that $I_\theta = \ell - |\ell_3|$. You may use $\int_0^\pi ds/(1 - a^2 \cos^2 s) = \pi/\sqrt{1 - a^2}$ for $0 \leq a < 1$.

Exercise 6 (Bohr–Sommerfeld for three wells). The old quantum rule of Lecture 6's closing reading is $I = \hbar n$. (i) A particle in a box: $H = \frac{1}{2}p^2$ on $0 < q < L$, reflected elastically at the walls. This is not a smooth Hamiltonian; take the orbit at energy E to bound the rectangle $0 \leq q \leq L$, $|p| \leq \sqrt{2E}$, and I to be its area over 2π . Find $I(E)$, check $T = 2\pi dI/dE$ against the time the particle takes to go across and back, and find the energies the rule allows. (ii) Find the energies the rule allows for the oscillator of Lecture 6, and for the Morse well of Exercise 1 (in terms of n with $\hbar n < \sqrt{2}$). (iii) The Schrödinger equation gives (cited, not derived here): for the box $E_n = \pi^2 \hbar^2 n^2 / (2L^2)$, $n = 1, 2, \dots$; for the oscillator $E_n = \hbar \nu (n + \frac{1}{2})$, $n = 0, 1, \dots$; for the Morse well $E_n = \sqrt{2} \hbar (n + \frac{1}{2}) - \frac{1}{2} \hbar^2 (n + \frac{1}{2})^2$ for the integers $n \geq 0$ with $\hbar(n + \frac{1}{2}) < \sqrt{2}$. For which of the three does the rule give these levels exactly, and for which does replacing n by $n + \frac{1}{2}$ in it give them exactly?

Exercise 7 ★. The Laplace–Runge–Lenz vector. On the phase space of $q \in \mathbb{R}^3 \setminus \{0\}$, $p \in \mathbb{R}^3$, let $H = \frac{1}{2}|p|^2 - \alpha/|q|$ with $\alpha > 0$, $\ell = q \times p$ (Sheet 2's L), and

$$A = p \times \ell - \alpha \frac{q}{|q|}.$$

Use $\{q_i, p_j\} = \delta_{ij}$, $\{q_i, q_j\} = \{p_i, p_j\} = 0$ and $\{\ell_i, \ell_j\} = \sum_k \epsilon_{ijk} \ell_k$ (Sheet 2, Exercise 1). (i) Show that $\{A_i, H\} = 0$ for $i = 1, 2, 3$. (ii) Show that $\ell \cdot A = 0$ and $|A|^2 = \alpha^2 + 2H|\ell|^2$. (iii) Show that $\{\ell_i, A_j\} = \sum_k \epsilon_{ijk} A_k$. (iv) Show that $\{A_i, A_j\} = -2H \sum_k \epsilon_{ijk} \ell_k$.